

Integrative metabolomics and machine learning reveal the toxic mechanisms of elaidic acid in polycystic ovary syndrome

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ABSTRACT

Background: Polycystic ovary syndrome (PCOS) is characterized by reproductive dysfunction and metabolic disturbances, including obesity, insulin resistance, and dyslipidemia. Dietary trans fatty acids, particularly elaidic acid (EA), have been implicated in metabolic disorders, but their role in PCOS pathogenesis remains unclear.

Methods: A letrozole combined with high-fat diet induced PCOS rat model was established to evaluate EA accumulation using serum metabolomics. We identified common genes linked to EA exposure and PCOS using WGCNA and differential expression analysis. Machine-learning models were applied to screen key genes. GSVA and CIBERSORT were used to assess pathway activity and immune infiltration. In vitro, human granulosa-like KGN cells were exposed to EA (0–50 μ M) to assess direct ovarian-cell responses, including cell viability, apoptosis, and inflammatory and key-gene mRNA expression. Molecular docking and molecular dynamics simulations examined EA-protein interactions.

Results: EA levels were markedly elevated in PCOS rats and accompanied by ovarian lipid deposition and metabolic dysregulation. Integrated analyses identified five key EA and PCOS related genes (KDM5A, CDK5, MTHFD1, NDUFB11, POLR1C). Machine-learning models confirmed their predictive value. GSVA revealed enrichment of inflammatory pathways such as JAK/STAT and Notch, while immune-cell profiling showed altered NK cells and macrophage subsets. In vitro, EA exposure significantly reduced KGN cell viability at higher concentrations and increased apoptotic fractions, accompanied by upregulation of IL6 and TNF- α and a modest increase in IL1B; EA also increased KDM5A expression while decreasing CDK5, MTHFD1, NDUFB11, and POLR1C, consistent with the direction of the integrative predictions.

Conclusions: EA accumulates in PCOS under high-fat dietary exposure and may exacerbate ovarian dysfunction by disrupting energy metabolism and promoting inflammatory activation, with concordant changes observed across serum metabolomics, transcriptomic integration, immune-infiltration analyses, and in vitro cell responses. These findings highlight EA as a potentially important dietary-derived toxic metabolic factor contributing to PCOS pathophysiology.

1. Introduction

Polycystic ovary syndrome (PCOS) is a common endocrine and metabolic disorder affecting women of reproductive age, with a prevalence of approximately 6–15%. Its main clinical features include hyperandrogenism, ovulatory dysfunction, and polycystic ovarian morphology. In addition to reproductive abnormalities, patients with

PCOS often present with metabolic syndrome features such as obesity, insulin resistance, and dyslipidemia, all of which contribute to an increased long-term risk of cardiovascular and metabolic diseases [1].

Dietary factors, particularly high-fat intake, are believed to play an important role in the onset and progression of PCOS [2]. Excessive lipid consumption can interfere with ovarian steroidogenesis [3], affect follicular development [4], and exacerbate insulin resistance [5]. Recent

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studies have shown that metabolic products derived from high-fat diets can directly regulate energy metabolism and signal transduction in ovarian cells, suggesting that lipid metabolites act as critical mediators linking nutritional composition to ovarian function [6]. Among these metabolites, elaidic acid (EA), the main industrial trans isomer of oleic acid, has attracted increasing attention due to its widespread presence in hydrogenated oils and processed foods. Previous studies have demonstrated that EA can induce systemic inflammatory responses [7], dyslipidemia, and endothelial dysfunction [8]. Its trans configuration reduces membrane fluidity and alters lipid signaling, thereby impairing metabolic homeostasis [9].

2. Methods

2.1. Experimental animals

Female Sprague-Dawley rats (3 weeks old, license No. SCXK [Jing] 2021-0011) were purchased from Charles River Laboratories. All animals were housed in an SPF barrier facility under controlled environmental conditions (temperature $22 \pm 2^\circ\text{C}$, humidity $55 \pm 10\%$) with a 12-h light/dark cycle. The study protocol was reviewed and approved by the Animal Ethics Committee of Nanjing University of Chinese Medicine (Approval No. 202204A054). After a three-day acclimation period, the rats were randomly assigned to the control group or the model group ($n = 6$ per group).

During the experiment, rats in the control group received standard chow and 1% sodium carboxymethyl cellulose (CMC-Na) by oral gavage. Rats in the model group were fed a high-fat diet (Research Diets, New Jersey, USA) and were administered letrozole at 1 mg/kg/day for 28 consecutive days, dissolved in 1% CMC-Na and given by oral gavage.

Body weight was recorded daily throughout the experiment. At the end of the study, all rats were fasted for 12 h and then euthanized under isoflurane anesthesia. Ovarian tissues were collected, with a portion fixed in paraformaldehyde and another portion fixed in commercial electron microscopy fixative. Blood samples were obtained from the abdominal aorta and centrifuged at 3000 rpm for 15 min at 4°C to prepare serum for subsequent analyses.

2.2. Assessment of estrous cycle

The estrous cycle of each rat was monitored daily by cytological examination of vaginal smears under a light microscope. The phase of the cycle was determined based on the relative proportions of leukocytes, cornified epithelial cells, and nucleated epithelial cells observed in the smears.

- Diestrus (D): dominated by small round leukocytes with sparse epithelial cells.
- Proestrus (P): characterized mainly by nucleated epithelial cells with minimal keratinization.
- Estrus (E): marked by abundant cornified (keratinized) epithelial cells and an absence of leukocytes.
- Metestrus (M): a mixed population of leukocytes and cornified epithelial cells, reflecting the transition phase.

2.3. ELISA

Serum levels of luteinizing hormone (LH), follicle-stimulating hormone (FSH), and testosterone (T) were quantified using commercial ELISA kits. All assays were performed in accordance with the manufacturer's protocols, and each sample ($n = 6$) was analyzed in duplicate to ensure measurement accuracy.

2.4. Transmission electron microscopy (TEM)

After collection, ovarian tissues were immediately placed in pre-

cooled normal saline and rinsed three times with pre-chilled PBS on ice. The tissues were cut into approximately 1 mm^3 pieces and fixed sequentially in 2.5 % glutaraldehyde at 4°C for 2–4 h and 1 % osmium tetroxide at 4°C for 1–2 h. Following PBS washes, the samples were dehydrated through a graded ethanol and acetone series, embedded in epoxy resin, and polymerized. Ultrathin sections (60–80 nm) were stained with uranyl acetate and lead citrate, and images were acquired using a transmission electron microscope.

2.5. Ovarian hematoxylin and eosin (H&E) staining

One ovary from each rat was collected, and surrounding adipose tissue was carefully removed. The tissue was immediately fixed in 4 % paraformaldehyde for 24 h. After graded ethanol dehydration and xylene clearing, the samples were embedded in paraffin and sectioned at a thickness of $4\text{ }\mu\text{m}$. The sections were deparaffinized, rehydrated, and stained with hematoxylin for 5 min. After rinsing in running water, differentiation in acid alcohol, and bluing, the sections were counterstained with eosin for 2 min, followed by dehydration in 70 %, 80 %, 95 %, and 100 % ethanol. After xylene clearing, the sections were mounted with neutral resin. Ovarian morphology and follicular development were observed under a light microscope.

2.6. Analysis of serum metabolic profiles

Serum metabolomic profiling was performed using an untargeted UHPLC-MS/MS platform. Briefly, 100 μL of serum from each sample ($n = 6$ per group) was extracted with 400 μL of pre-chilled methanol containing isotopically labeled internal standards. After vortexing, sonication, protein precipitation at -40°C , and centrifugation (12,000 rpm, 15 min, 4°C), the supernatant was collected for analysis. A pooled quality control (QC) sample was prepared by mixing equal aliquots from all samples and injected periodically to monitor system stability.

Chromatographic separation was carried out on a Vanquish UHPLC system (Thermo Fisher Scientific) equipped with an HSS T3 column ($2.1 \times 100\text{ mm}$, $1.8\text{ }\mu\text{m}$), using 5 mM ammonium acetate with 5 mM acetic acid in water (A) and acetonitrile (B) as the mobile phases. The injection volume was 2 μL , and the autosampler temperature was maintained at 4°C .

Mass spectrometric analysis was performed on an Orbitrap Exploris 120 mass spectrometer (Thermo Fisher Scientific) operating in both positive and negative electrospray ionization modes. Full MS data were acquired at a resolution of 60,000, and data-dependent MS/MS spectra were acquired at a resolution of 15,000.

Raw LC-MS data were converted to mzXML format using ProteoWizard and processed in R (XCMS package) for peak detection, alignment, and integration. Features with relative standard deviation (RSD) $> 30\%$ in QC samples or missing values in more than 50% of samples per group were removed. Remaining missing values were imputed using half of the minimum detected value, and peak intensities were normalized to internal standards prior to statistical analysis.

Metabolite annotation was performed by matching accurate mass, retention time, and MS/MS fragmentation patterns against an in-house spectral library (BiotreeDB). To reduce false-positive identifications, only metabolites with MS/MS-supported annotations were retained. Given the structural similarity among unsaturated fatty acid isomers, EA was further confirmed by its characteristic secondary MS/MS fragmentation pattern and consistent retention time across samples. Relative abundance of EA was quantified based on normalized peak areas (Fig. 1L).

Multivariate statistical analyses, including principal component analysis (PCA) and orthogonal partial least squares-discriminant analysis (OPLS-DA), were performed using SIMCA (v16.0.2). Model robustness was evaluated by 7-fold cross-validation and 200 permutation tests. Differential metabolites were defined based on Student's *t*-test

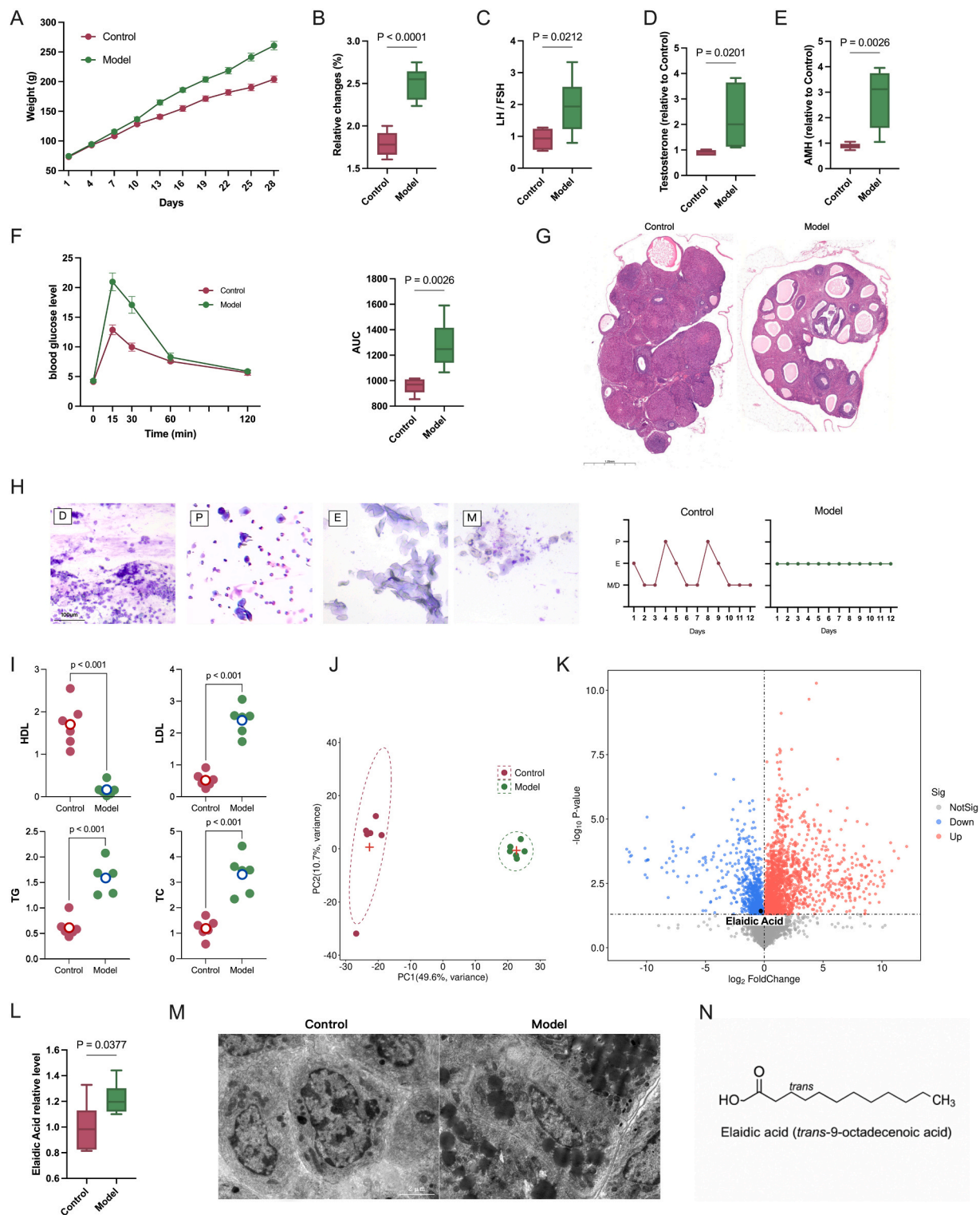


Fig. 1. Phenotypic, metabolic, and ultrastructural features of PCOS model rats. (A) Longitudinal body weight curves of control and model rats (n = 6). (B) Relative weight gain across the experimental period (n = 6). (C) Serum LH/FSH ratio, showing a significant increase in the model group (n = 6). (D) Serum testosterone levels (n = 6) and (E) AMH levels (n = 6), both elevated in the model group. (F) Intraperitoneal glucose tolerance test (IPGTT) and corresponding area under the curve (AUC), indicating impaired glucose clearance in the model rats (n = 6). (G) Representative ovarian H&E staining demonstrating typical polycystic morphology, including multiple cystic follicles, diminished corpora lutea, and disrupted follicular development. (H) Vaginal cytology analysis showing representative smears and estrous cycle patterns (n = 6). Model rats exhibited disrupted cycling with prolonged diestrus and loss of normal periodicity. (I) Serum lipid profiles demonstrating decreased HDL and increased LDL, TC, and TG levels in the model group (n = 6). (J) PCA score plot illustrating clear separation of serum metabolomic signatures between groups (n = 6). (K) Volcano plot highlighting elaidic acid as one of the most significantly upregulated metabolites in model rats. (L) Relative abundance of elaidic acid in serum samples. (M) Transmission electron micrographs of ovarian tissue showing marked accumulation of intracellular lipid droplets and mitochondrial structural abnormalities in model animals. (N) Chemical structure of elaidic acid (trans-9-octadecenoic acid).

($p < 0.05$) combined with variable importance in projection (VIP > 1), and pathway enrichment analysis was conducted using KEGG annotations.

2.7. Cell culture and treatment

KGN cells were cultured in DMEM/F12 (Keygen Biotech) supplemented with 10% charcoal-stripped fetal bovine serum (Gibco) and 1% penicillin-streptomycin-neomycin antibiotic mixture (Gibco). Cells were maintained at 37°C in a humidified 5% CO₂ incubator (Thermo Fisher Scientific) and routinely passaged. To investigate the effects of elaidic acid, cells were treated with varying concentrations (0–50 µM) of the compound for 24 h. Elaidic acid was dissolved in dimethyl sulfoxide (DMSO), and an equal volume of DMSO was added to the control groups.

2.8. Cell viability assay

Cell viability was assessed using a Cell Counting Kit-8 (CCK-8; TargemoI). Briefly, cells were seeded into 96-well plates at 5000 cells per well. After 24 h of culture, 10 µl of CCK-8 reagent was added to each well. Following a 1-hour incubation at 37°C, the absorbance at 450 nm was recorded using a microplate reader. Each experiment was performed with three technical replicates and independently repeated three times.

2.9. Quantitative real-time PCR (qPCR)

Total RNA was extracted using the RNAeasy™ kit (Beyotime Biotechnology) and reverse transcribed into cDNA with a SuperMix reverse transcription system (Vazyme). Quantitative PCR was subsequently carried out on a Quantstudio 5 real-time PCR system (Thermo Fisher Scientific) using SYBR Green PCR Master Mix (Vazyme). The expression level of the target genes was normalized to that of β-actin. Primer sequences are listed in Table 1.

2.10. Cell apoptosis analysis by flow cytometry

Apoptosis was evaluated by flow cytometry using an Annexin V-APC/7-AAD Apoptosis Kit (Catalog No. AT105; Lianke Bio) according to the manufacturer's instructions. Following treatment, cells were harvested, washed with cold PBS, and resuspended in 1 × binding buffer. Subsequently, the cells were stained with Annexin V-APC and 7-AAD for 15 min at room temperature in the dark. Samples were diluted with additional binding buffer and analyzed immediately using a beckman coulter flow cytometer. Apoptotic populations were determined as follows: Annexin V–/7-AAD– (viable), Annexin V+ /7-AAD– (early apoptotic), Annexin V+ /7-AAD+ (late apoptotic), and Annexin V–/7-AAD+ (necrotic). Data analysis was performed using FlowJo software.

2.11. Data acquisition and integration

Acquisition and Integration of Potential Targets of Elaidic Acid: To systematically identify the potential molecular targets of elaidic acid

(EA), we retrieved predicted target genes from three independent databases, including ChEMBL, SEA (Similarity Ensemble Approach), and SwissTargetPrediction. All predicted targets were merged to generate a unified EA-related target set. A Venn diagram was used to visualize the overlap among the three databases and to determine the shared and unique EA-associated genes.

Acquisition and Batch Correction of PCOS-Related Transcriptomic Data: Multiple publicly available PCOS transcriptomic datasets were downloaded from the GEO database. To reduce experimental heterogeneity among studies, batch correction was performed using the ComBat function in the sva package. Differences in sample distribution before and after correction were assessed through boxplots and principal component analysis (PCA) to confirm effective removal of batch effects and improved dataset uniformity.

2.12. Differential expression analysis and WGCNA

After data integration, differential expression analysis (DEG) was performed between PCOS and control groups using the limma package. Significant DEGs were identified based on an absolute log₂ fold change greater than one and a false discovery rate (FDR) less than 0.05.

Weighted gene co-expression network analysis (WGCNA) was then conducted to construct a co-expression network. The analysis included: (1) calculation of the topological overlap matrix; (2) identification of co-expression modules using a dynamic tree-cutting algorithm; (3) evaluation of correlations between module eigengenes and PCOS phenotype to determine key modules; (4) assessment of module membership (MM) and gene significance (GS) to identify core genes within key modules. DEGs were intersected with genes from the key WGCNA modules to generate a set of PCOS-related candidate genes.

2.13. Identification of shared EA-PCOS genes and functional enrichment

The EA target set was intersected with PCOS-related candidate genes to identify shared EA-PCOS genes. Protein-protein interaction (PPI) networks were constructed using the STRING database and visualized in Cytoscape for identification of central interaction nodes.

Functional enrichment analysis of shared genes was performed using the clusterProfiler package, including Gene Ontology (GO) enrichment for biological process (BP), cellular component (CC), and molecular function (MF), as well as KEGG pathway enrichment. These analyses were used to explore potential biological functions and pathways associated with EA-PCOS shared genes.

2.14. Machine learning model construction and key gene selection

To evaluate the predictive value of EA-PCOS shared genes, four machine learning models were constructed: generalized linear model (GLM), support vector machine (SVM), random forest (RF), and XGBoost (XGB). Ten-fold cross-validation was used to assess model stability. The performance of each model was evaluated by residual distribution, receiver operating characteristic (ROC) curves, and the area under the curve (AUC). Feature importance rankings were used to identify key predictors.

The GLM model with optimal performance was further used to generate a PCOS risk prediction nomogram. Model calibration was assessed using calibration curves, and clinical utility was evaluated using decision curve analysis (DCA). External validation was performed with two GEO datasets, GSE43322 and GSE43264.

2.15. GSVA pathway activity and immune cell infiltration analysis

To explore potential biological functions of the key genes, gene set variation analysis (GSVA) was performed using KEGG gene sets to estimate pathway activity differences between PCOS and control groups.

Immune cell infiltration was assessed using the CIBERSORT

Table 1
Primer sequence.

Gene name	Forward primer (5'→3')	Reverse primer (5'→3')
β-actin	CTCGCCCTTGCCGATCC	ATCCTTCTGACCCATGCCC
IL-1β	CTGTACTCTGCTGCGTGT	GGGAAGTGGGAGACTCAAA
IL-6	TGAGGAGACTTGCTGGTGAA	CAGCTCTGGCTTGTCTCTCAC
TNF-α	TATCCTGGGGACCCAATGT	AAAAGAAGGCACAGAGGCCA
CDK5	GGAAGGCACCTACGGAATG	GGCACACCTCATCATCGTC
KDM5A	ACTTGAACGGGCTTAGTGGG	TCCCAACTCGGCTCAAAGAC
MTHFD1	GGCTTTTGATGCCGTGAAGT	GCTTTGTGTGAGCTTCGGG
NDUFB11	ACGTCTGGAACATGCGACTT	GGCAGATAGGCCACAAAGGT
POLR1C	GGGAATTGACGCAGCCATTG	CCCAGACGGTGAGCAAGAAT

algorithm, which quantifies the proportions of 22 immune cell subsets. Differences in immune composition between PCOS and control groups were compared, and Spearman correlation analysis was conducted to evaluate the associations between the five key genes (CDK5, KDM5A, MTHFD1, NDUFB11, POLR1C) and immune cell infiltration levels.

2.16. Molecular docking and molecular dynamics simulation

The three-dimensional structure of EA was obtained from PubChem and converted to PDBQT format using AutoDockTools. Crystal structures of the five key proteins (KDM5A, POLR1C, MTHFD1, CDK5, NDUFB11) were downloaded from the Protein Data Bank and pre-processed by removing water molecules, adding hydrogen atoms, and assigning partial charges. Molecular docking was performed using AutoDock Vina, and binding energies and key interacting residues were analyzed.

Based on docking results, the EA-POLR1C complex with the lowest binding energy was selected for molecular dynamics simulation. Simulations were conducted using GROMACS 2022. The GAFF2 force field was applied to EA, and the amber14sb force field was used for the protein. The pdb2gmx module was used for initial structure pre-processing. The complex was placed in a TIP3P water box with a 1 nm margin, and Na⁺/Cl⁻ ions were added to neutralize the system using gmx genion.

Long-range electrostatic interactions were calculated using the Particle Mesh Ewald method with a cutoff of 1.0 nm. All covalent bonds were constrained using the LINCS (or SHAKE) algorithm, and the time step was set to 1 fs.

Energy minimization was performed using 3000 steps of steepest descent followed by 2000 steps of conjugate gradient optimization. Equilibration included 100 ps of NVT and 100 ps of NPT ensembles at 310 K and 1 bar. The production phase consisted of a 100 ns NPT simulation.

Post-simulation analyses included root mean square deviation (RMSD), root mean square fluctuation (RMSF), hydrogen bond counts, radius of gyration (Rg), solvent-accessible surface area (SASA), and free energy landscape (FEL), all computed through standard GROMACS tools.

2.17. Statistical analysis

All analyses were conducted using R (v1.1.2; data of use: 2023.12.7). All results are expressed as mean $\pm$ SEM from at least three independent experiments. Comparisons between two groups were conducted using a Student's *t*-test, and one-way ANOVA was applied for multiple-group analyses. Statistical significance was defined as $P < 0.05$.

All statistical analyses were performed using R software (version 4.3.1). Continuous data are expressed as mean $\pm$ standard error of the mean (SEM). Comparisons between two groups were conducted using the Wilcoxon signed-rank test, yielding two-tailed *p*-values. Differentially expressed genes, pathway enrichment, and correlation analyses underwent multiple testing correction using the FDR method implemented via R's *p.adjust* function. Unless otherwise specified, corrected *p*-values (FDR) < 0.05 were considered statistically significant. Experimental validation results are presented as the mean $\pm$ standard error of the mean (SEM) from at least three independent experiments. Statistical significance for experimental data was defined as $p < 0.05$.

3. Results

3.1. The PCOS rat model exhibits metabolic disturbances and accumulation of laidic acid

We established a PCOS rat model using letrozole combined with a high-fat diet and evaluated its characteristics at the phenotypic, metabolomic, and ovarian structural levels. Compared with the control

group ($n = 6$), the model rats showed continuous body weight gain during the experiment (Fig. 1A), an increased relative weight change (Fig. 1B), a higher serum LH/FSH ratio (Fig. 1C), elevated testosterone levels (Fig. 1D), and increased AMH concentrations (Fig. 1E). The intraperitoneal glucose tolerance test (IPGTT) indicated impaired glucose clearance in the model group, with a significantly higher AUC value (Fig. 1F). In addition, the model rats exhibited disrupted estrous cycles (Fig. 1H), and ovarian H&E staining showed typical polycystic changes, including an increased number of cystic follicles and a marked reduction in corpora lutea (Fig. 1G), confirming that the PCOS model was successfully induced.

PCOS rats also showed pronounced lipid metabolic abnormalities, including reduced HDL and elevated LDL, TC, and TG levels (Fig. 1I). To assess the metabolic disturbances associated with the high-fat diet, we performed untargeted metabolomics. Untargeted metabolomic analysis identified a total of 872 metabolites after data preprocessing and quality control. 217 metabolites were identified as differentially expressed between the control and model groups. The PCA plot demonstrated clear separation between the two groups (Fig. 1J), indicating substantial differences in metabolic profiles. Among the differential metabolites, EA showed a particularly prominent increase (Fig. 1K-L), which is consistent with its origin from a high-fat diet and may reflect systemic enrichment of industrial trans fatty acids.

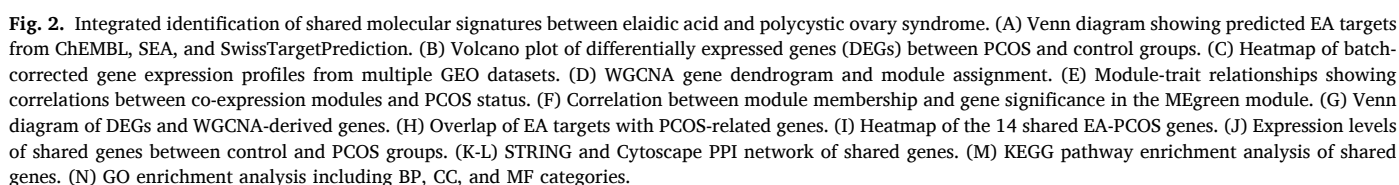
Previous studies [10] reported that EA preferentially accumulates in white adipose tissue and acts directly on cells. Transmission electron microscopy further supported this observation: ovarian cells in the model group displayed more lipid droplet deposition and mitochondrial abnormalities compared with the control group (Fig. 1M), suggesting that EA may influence local lipid metabolism within the ovary. The chemical structure of EA is shown in Fig. 1N.

3.2. Identification of genes between EA and PCOS

To further identify the potential molecular targets of EA in PCOS, we applied a multi-database prediction strategy and systematically extracted EA-related targets from the ChEMBL, SEA, and SwissTargetPrediction platforms. The Venn diagram (Fig. 2A) shows partial overlap among the databases, with 3 genes shared across all three sources. After merging, a total of 933 potential EA targets were obtained.

To improve the accuracy of integrating multiple public transcriptome datasets, we compiled the gene expression matrices from several GEO datasets and evaluated batch effects before and after correction. As shown in Supplementary Figure 1, before batch correction, the expression distributions differed markedly across studies, with uneven boxplots and distinct separation of samples from different datasets in the PCA plot, indicating substantial batch effects. After batch correction, the expression distributions became more consistent, and samples from different sources clustered more closely in the PCA space, demonstrating that the batch effects were effectively removed and that the data were suitable for subsequent differential analysis and network exploration (Fig. 2C).

Differential expression analysis identified 67 significantly altered genes between PCOS and controls (Fig. 2B). To further explore gene co-expression patterns, we performed weighted gene co-expression network analysis (WGCNA). The clustering dendrogram and module assignment indicated several co-expression modules (Fig. 2D). Module-trait relationship analysis showed that the eigengene of the green module (MEgreen) exhibited a weak positive correlation with PCOS status, which did not reach statistical significance ($r = 0.05$, $p = 0.5$) (Fig. 2E). Given that module eigengenes represent averaged expression profiles and may underestimate trait-relevant signals in heterogeneous modules, we further evaluated gene-level associations within each module by examining the relationship between module membership (MM) and gene significance (GS) for PCOS. Notably, within the green module, MM was significantly and positively correlated with GS ($\text{cor} = 0.38$, $p = 1.2 \times 10^{-9}$) (Fig. 2F), indicating that genes with higher



intramodular connectivity tend to show stronger associations with the PCOS phenotype. This consistent internal structure suggests that the green module contains PCOS-relevant co-expression signals at the gene level. Therefore, the green module was considered a module of interest for subsequent downstream analyses.

Combining DEGs and WGCNA results (Fig. 2G), 7 intersection genes

were identified. To uncover the shared molecular basis between EA and PCOS, we intersected the 933 EA targets with the 185 PCOS-related genes, yielding 14 common genes (Fig. 2H). Heatmap visualization (Fig. 2I) and differential expression analysis (Fig. 2J) confirmed distinct expression patterns of these genes between groups.

Protein-protein interaction (PPI) network analysis using STRING and

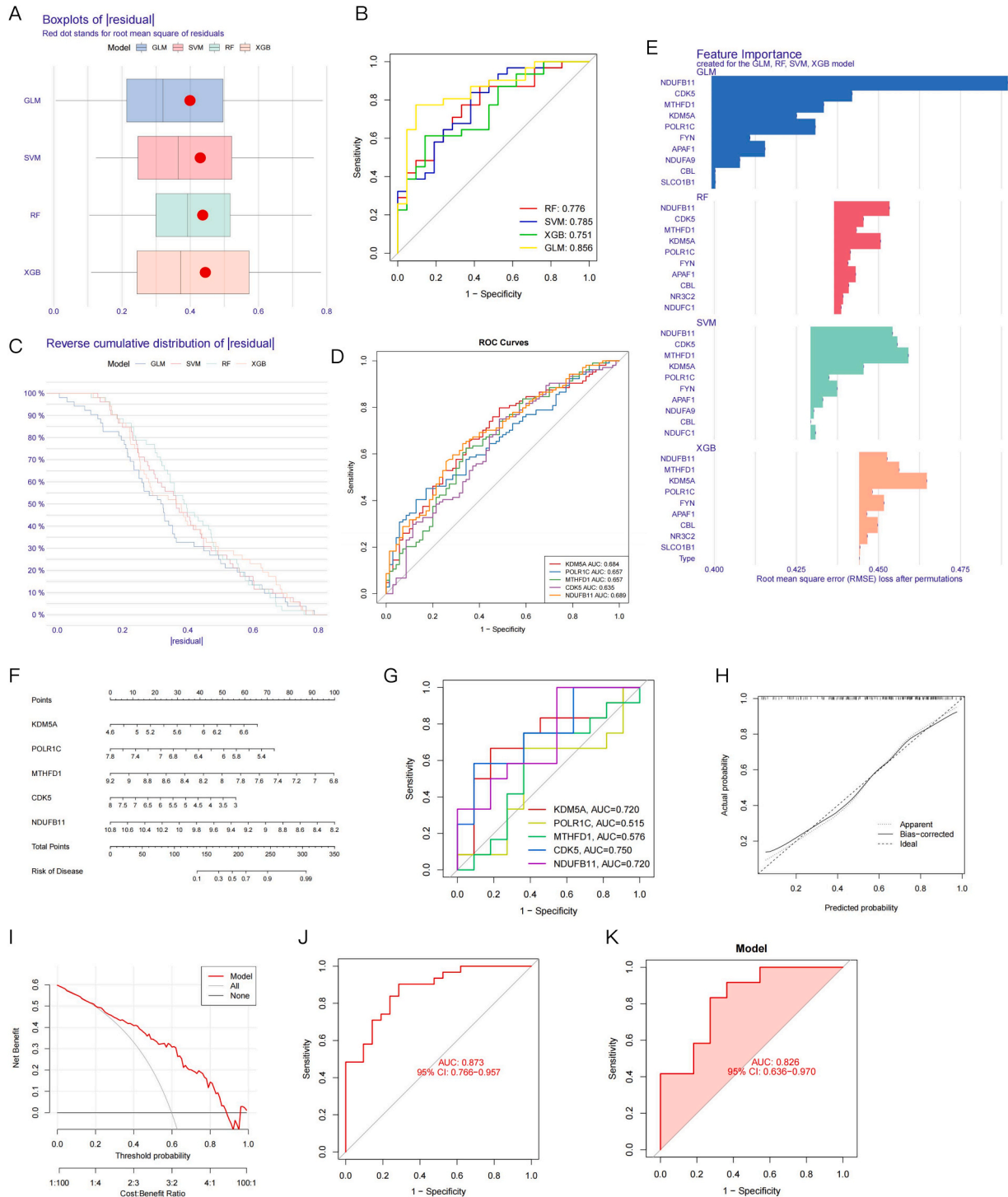


Fig. 3. Performance evaluation of machine learning models and identification of key EA-PCOS predictive genes. (A) Boxplots of residuals for GLM, SVM, RF, and XGB models. (B) ROC curves comparing discriminative performance of the four models. (C) Reverse cumulative distribution of residuals. (D) Single-gene ROC curves for the 14 shared EA-PCOS genes. (E) Feature importance ranking across GLM, RF, SVM, and XGB models. (F) Nomogram constructed from the GLM model using five key genes. (G) Independent ROC analysis of five core genes (CDK5, KDM5A, MTHFD1, NDUFB11, POLR1C). (H) Calibration curve evaluating agreement between predicted and observed probabilities. (I) Decision curve analysis demonstrating net clinical benefit. (J) Internal validation ROC curve. (K) External validation ROC curve using GSE43322 and GSE43264 datasets.

Cytoscape revealed dense interactions among the 14 genes (Fig. 2K-L). Several genes, including APAF1, CBL, CDK5, FYN, KDM5A, MTHFD1, NDUFB11, and POLR1C, occupied central positions, implicating energy metabolism, signal regulation, and stress responses.

Functional enrichment further clarified the biological roles of the EA-PCOS shared genes. KEGG analysis indicated enrichment in neurodegenerative disease pathways, oxidative phosphorylation, and nonalcoholic fatty liver disease (Fig. 2M). GO analysis showed that BP was dominated by purine and nucleotide biosynthetic processes; CC was enriched in the mitochondrial respiratory chain complex; MF was focused on oxidoreductase activity and electron transfer (Fig. 2N). These findings indicate that EA may impair ovarian function in PCOS by disrupting purine metabolism and mitochondrial oxidative phosphorylation, which may lead to energy imbalance and increased oxidative

stress.

3.3. Machine learning models identify key EA-PCOS risk predictors

Based on the 14 shared EA-PCOS genes identified earlier, we constructed four machine learning models (GLM, SVM, RF, and XGB) to evaluate their predictive performance for PCOS risk. The residual plots (Fig. 3A) showed that GLM had the smallest and most concentrated residuals, indicating the lowest fitting error. SVM and RF exhibited moderate performance, while XGB displayed wider residual variation.

To compare the discriminative ability of each model, ROC curves were generated (Fig. 3B). The GLM model achieved the highest AUC (0.856), outperforming SVM (0.785), RF (0.776), and XGB (0.751), suggesting that GLM provided the strongest classification capability for

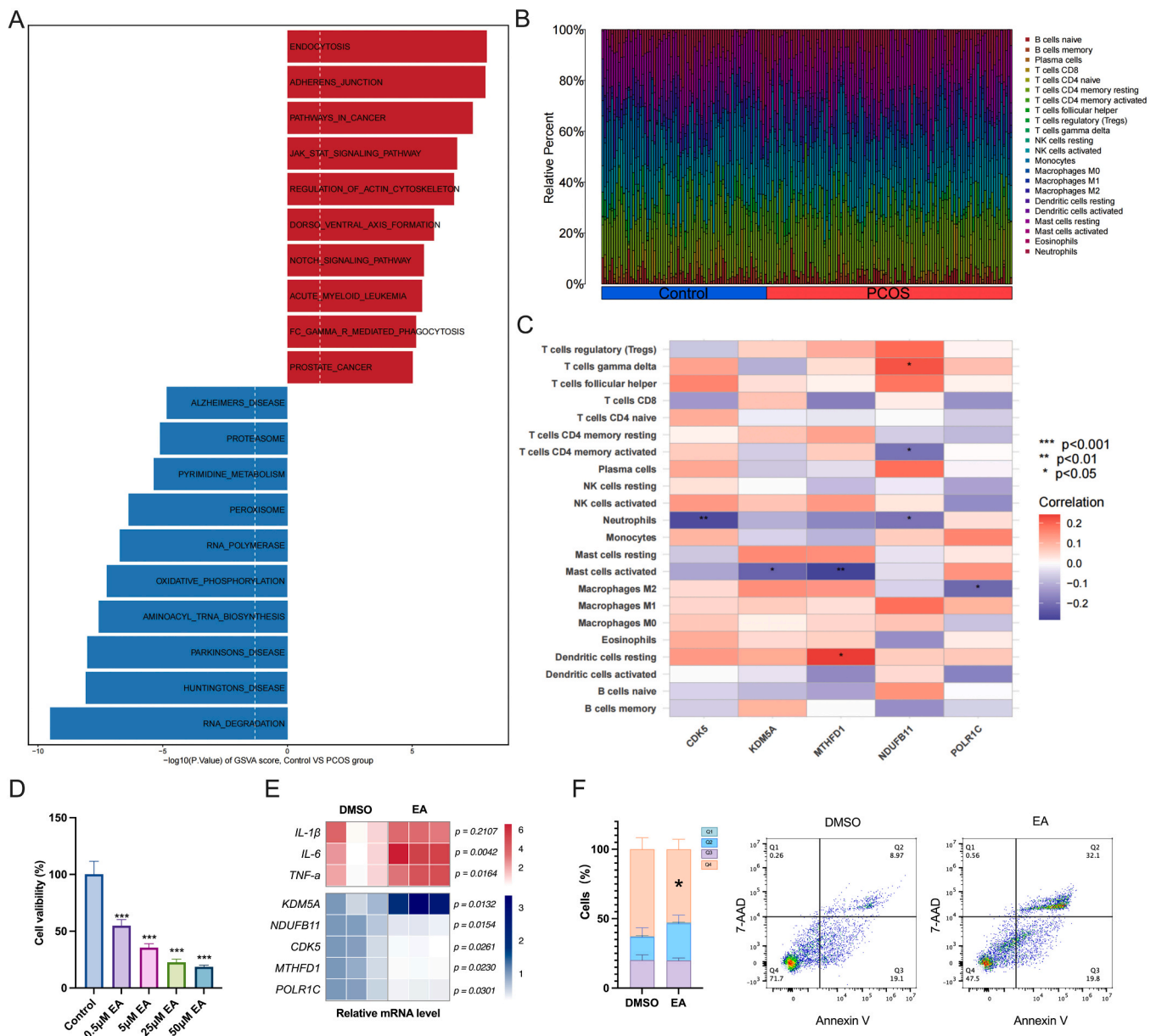


Fig. 4. GSEA functional enrichment and immune cell infiltration analysis in PCOS. (A) GSEA enrichment showing upregulated pathways in the PCOS group (red) and control group (blue). (B) CIBERSORT-estimated proportions of 22 immune cell subtypes in control and PCOS samples. (C) Correlation heatmap showing associations between five key genes (CDK5, KDM5A, MTHFD1, NDUFB11, POLR1C) and immune cell infiltration levels. (D) EA induced changes in KGN cell viability assessed by the CCK-8 assay. (E) Relative mRNA expression levels of inflammatory cytokines (IL6, TNF-α, and IL1B) and the five key genes identified by integrative bioinformatic and machine-learning analyses in KGN cells following EA treatment (50 μM). (F) Flow cytometric analysis of apoptosis in KGN cells after EA treatment (50 μM), evaluated by Annexin V/PI staining, showing changes in viable, early apoptotic, and late apoptotic cell populations.

PCOS risk. The reverse cumulative distribution of residuals also confirmed the superior stability of GLM (Fig. 3C).

Feature importance analysis across the four models (Fig. 3E) consistently highlighted NDUFB11, CDK5, MTHFD1, KDM5A, and POLR1C as key contributors. To further assess their individual discriminative power, single-gene ROC curves were plotted (Fig. 3D), among which NDUFB11 showed the highest AUC (0.689), suggesting its potential as a molecular marker for EA-mediated PCOS risk.

Based on the selected five genes, a GLM-based nomogram was constructed to estimate individual PCOS risk (Fig. 3F). Additional single-gene ROC analyses demonstrated that CDK5 (AUC = 0.750), KDM5A (AUC = 0.720), and NDUFB11 (AUC = 0.720) had good independent predictive ability (Fig. 3G).

The calibration curve (Fig. 3H) indicated good agreement between predicted and observed probabilities. Decision curve analysis (Fig. 3I) further showed that the GLM model yielded higher net benefit across a wide range of threshold probabilities compared with the “treat all” or “treat none” strategies.

Internal validation demonstrated excellent model performance (Fig. 3J, AUC = 0.873, 95% CI: 0.766–0.957). External validation using GSE43322 and GSE43264 (Fig. 3K) confirmed that the model maintained good predictive capability in independent datasets (AUC = 0.826, 95% CI: 0.636–0.970), supporting its generalizability.

3.4. EA exposure activates inflammatory signaling in granulosa cells

To further evaluate the potential biological functions of these genes, we performed GSVA analysis (Fig. 4A). The PCOS group was significantly enriched in inflammation- and signaling-related pathways, while the control group was enriched in metabolic pathways, including oxidative phosphorylation, pyrimidine metabolism, and RNA degradation. These results suggest that elaidic acid may promote the development and progression of PCOS by disrupting energy metabolism pathways and activating immune signaling.

To investigate the interaction between elaidic acid-related molecular features and the immune microenvironment in PCOS, we first used the CIBERSORT method to analyze immune cell infiltration (Fig. 4B). Compared with the control group, multiple immune cell subsets showed significant changes in the PCOS group, indicating immune state remodeling.

We then conducted correlation analysis between the expression levels of the five key genes (CDK5, KDM5A, MTHFD1, NDUFB11, POLR1C) and immune cell subsets (Fig. 4C). The results showed that these core genes were significantly correlated with multiple immune cell populations, especially NK cells, monocytes, dendritic cells, and M1/M2 macrophages, suggesting that these metabolism-related genes may be involved in immune regulation.

KGN cells were used to evaluate the direct effects of EA on ovarian cells. Cells were treated with increasing concentrations of EA (0–50 μ M). EA exposure resulted in a dose-dependent reduction in cell viability, with a significant inhibitory effect observed at 50 μ M (Fig. 4D). Analysis of inflammatory gene expression showed that EA treatment markedly increased the mRNA levels of IL6 and TNF- α , while IL1B exhibited a modest but detectable upregulation (Fig. 4E). In parallel, EA exposure significantly altered the expression of several genes identified in the integrative bioinformatic analyses: KDM5A expression was upregulated, whereas CDK5, MTHFD1, NDUFB11, and POLR1C were downregulated in KGN cells following EA treatment (Fig. 4E).

Flow cytometric analysis further demonstrated that EA induced apoptosis in KGN cells, characterized by a reduction in the viable cell population and a corresponding increase in early and late apoptotic fractions (Fig. 4F). These cellular changes indicate that EA directly disrupts granulosa cell survival and inflammatory homeostasis, which may contribute to impaired follicular function associated with PCOS.

3.5. Molecular docking and molecular dynamics analysis of EA

To explore potential molecular targets of EA, we performed molecular docking and molecular dynamics (MD) simulations as predictive analyses to nominate candidate EA-responsive proteins. We first carried out molecular docking for the five core targets (KDM5A, POLR1C, MTHFD1, CDK5, NDUFB11) and ranked the results based on the binding energy (BE). As shown in Fig. 5A, POLR1C exhibited the lowest binding energy with EA (−6.3 kcal/mol), followed by NDUFB11 (−6.2 kcal/mol), indicating that POLR1C has the strongest binding affinity and may serve as a primary binding target.

To further investigate the energetic stability of the binding conformation, we constructed the free energy landscape (FEL) of the EA-POLR1C complex. As shown in Fig. 5B, a pronounced deep energy well was observed, suggesting that the complex tends to form a low-energy and stable state during the simulation.

The binding mode analysis (Fig. 5C) showed that multiple interaction types were formed between EA and the POLR1C binding pocket. ALA202, LEU264, LEU212, PRO159, TYR210, VAL205, THR280, LEU197, LEU194, ALA198, MET201 and LEU209 contributed to extensive van der Waals interactions. GLY211 formed a carbon hydrogen bond, while LEU276 and ILE277 formed alkyl interactions, which together enhanced the binding stability.

To further evaluate the conformational stability of the complex in a dynamic environment, a 100 ns molecular dynamics simulation was performed. As shown in Fig. 5D–H, the RMSD of the complex reached stability after approximately 80 ns and remained within a fluctuation range of about 4.8 Å. The radius of gyration (Rg) and solvent accessible surface area (SASA) remained stable, and the number of hydrogen bonds fluctuated between 1 and 4, with most of the simulation maintaining around 2 hydrogen bonds. RMSF analysis (Figs. 5F and 5I) revealed low residue flexibility except in several exposed loop regions, indicating low overall flexibility and a stable complex structure.

In summary, EA is capable of stable binding and sustained dynamic interaction with POLR1C, suggesting that POLR1C may be an important target through which EA exerts its effects in PCOS [11].

4. Discussion

This study reveals the marked accumulation of elaidic acid in a rat model of PCOS, and by integrating network toxicology and machine learning results, it proposes potential mechanisms through which EA may influence ovarian function by disrupting energy metabolism and inflammatory signaling pathways.

Recent studies have shown that among the metabolic products of high-fat diets, trans fatty acids, particularly EA, have been implicated in the pathogenesis of metabolic disturbances. As the major industrial trans isomer of oleic acid, EA is widely present in hydrogenated oils and processed foods and may induce systemic inflammation, dyslipidemia, and endothelial dysfunction [11]. Previous reports have indicated that EA can trigger vascular endothelial inflammation, reduce nitric oxide (NO) production, and aggravate dyslipidemia [12]. Animal studies have also demonstrated that EA accumulates in white adipose tissue (WAT), inducing endoplasmic reticulum stress, insulin resistance, and lipid metabolic disorders [13]. In our metabolomics analysis, serum EA levels were significantly elevated in PCOS rats, accompanied by dyslipidemia, obesity, and other metabolic syndrome features. This pattern reflects trans fatty acid accumulation under long-term high-fat diet exposure and its profound disturbance of systemic metabolic homeostasis in PCOS.

At the molecular level, EA may affect ovarian cell function through pathways related to energy and nucleotide metabolism. Mitochondrial dysfunction is considered an important feature of ovarian metabolic disturbance in PCOS [14], and as a trans fatty acid, EA may impair membrane fluidity, alter respiratory chain complex activity, and promote ROS accumulation, leading to impaired cellular energy

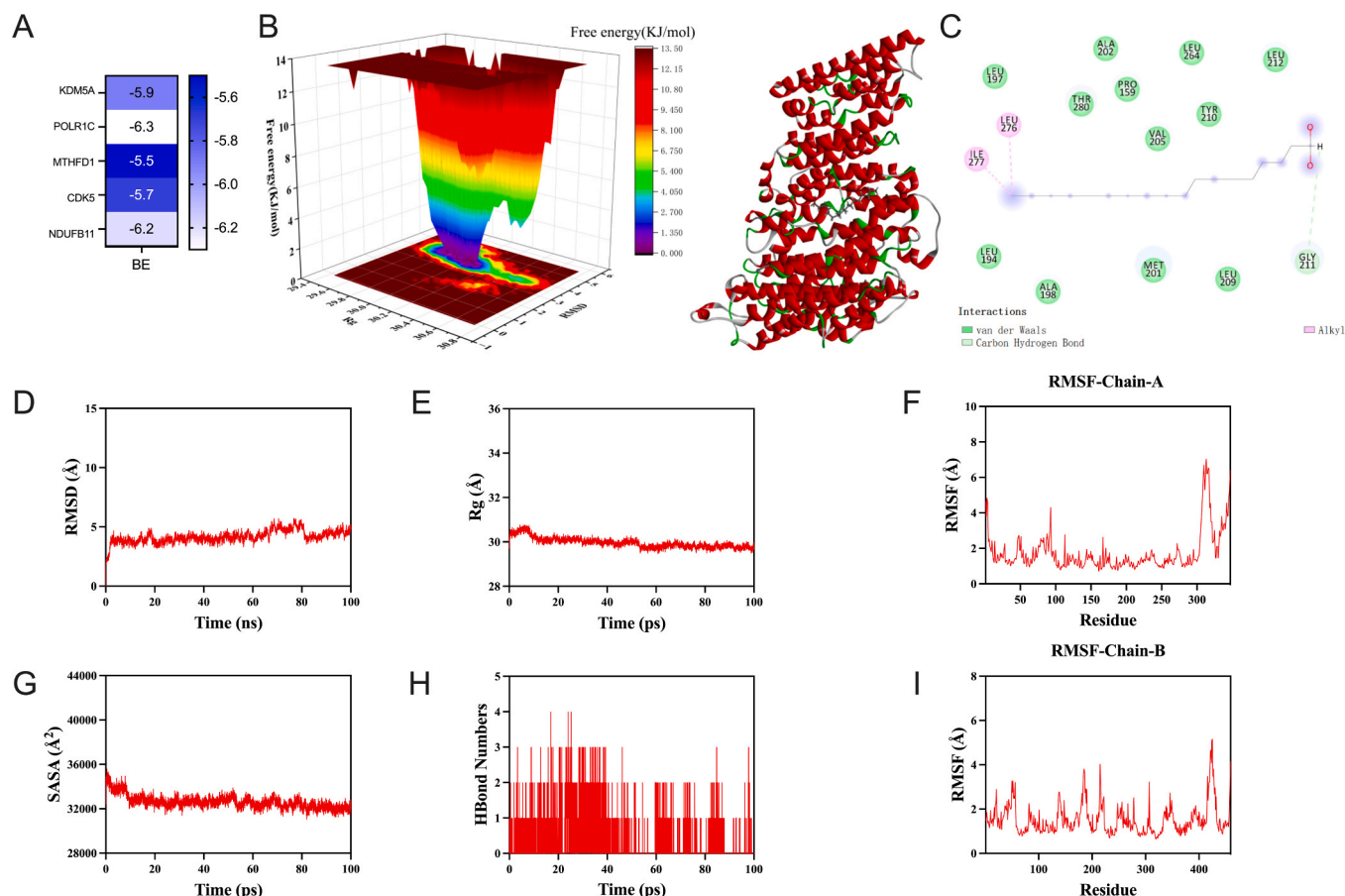


Fig. 5. Molecular docking and molecular dynamics analysis of EA binding to POLR1C. (A) Binding energy heatmap for EA docked with five core targets. (B) Free energy landscape (FEL) of the EA-POLR1C complex and the corresponding protein conformation. (C) Binding interaction map showing van der Waals interactions, carbon hydrogen bonds, and alkyl interactions between EA and POLR1C. (D) RMSD plot of the EA-POLR1C complex over 100 ns. (E) Radius of gyration (Rg) during the simulation. (F) RMSF of chain A residues. (G) Solvent accessible surface area (SASA) over the simulation time. (H) Number of hydrogen bonds formed during the simulation. (I) RMSF of chain B residues.

metabolism and increased apoptotic tendency [15]. Combined with the machine-learning-based gene screening results, this study identified several key genes involved in EA-related metabolic stress responses in PCOS. EA-related stress responses appear not to be limited to membrane lipids or classical nuclear receptor pathways but may also involve ribosome biogenesis-related signaling.

Existing evidence indicates a strong association between elaidic acid and elevated inflammatory markers. Diets enriched in industrial trans fatty acids increase circulating levels of C-reactive protein (CRP), fibrinogen, and interleukin-6, reflecting a state of chronic low-grade inflammation [16]. PCOS is recognized as a metabolic disorder accompanied by chronic low-grade inflammation, characterized by elevated pro-inflammatory cytokines and aberrant immune cell infiltration [17]. Our GSVA and immune-infiltration analyses further showed significant activation of inflammation-related pathways in the PCOS model, suggesting a central role for immune remodeling in its pathogenesis. GSVA revealed significant enrichment of inflammatory and signaling pathways—including JAK/STAT and Notch—in the PCOS group, indicating that EA may promote PCOS development and progression by disturbing energy metabolism and activating immune signaling. Previous studies have shown that the JAK/STAT pathway participates in abnormal follicle development and steroidogenesis in PCOS [18], while Notch signaling plays essential roles in follicular assembly and granulosa cell proliferation [19]. Ovarian ER stress can activate Notch2 signaling, disrupt granulosa cell function, and contribute to ovulatory dysfunction [20]; Notch-related molecules are abnormally expressed in cumulus cells of PCOS patients and are associated with oocyte and embryo

quality [21]. These findings suggest that dysregulated Notch signaling may be a critical contributor to impaired follicular development in PCOS.

CIBERSORT analysis further revealed significant alterations in multiple immune cell subsets in PCOS, including NK cells, monocytes, dendritic cells, and M1/M2 macrophages, indicating typical immune-state remodeling. The key metabolism-related genes were strongly correlated with the infiltration levels of these immune cell populations, especially NK cells and macrophage subsets. This suggests that EA may reshape the ovarian inflammatory microenvironment and aggravate PCOS phenotypes by modulating immune-cell activation and distribution through metabolic signaling pathways. KDM5A, an epigenetic regulator, has been shown to promote M1 macrophage polarization and enhance pro-inflammatory gene expression, with its activation closely associated with upregulated NF- κ B signaling [22,23]. POLR1C, a shared subunit of RNA polymerase I/III, participates in rRNA transcription and ribosome biogenesis [24]. Impaired Pol I/III function can induce canonical nucleolar stress and activate p53 signaling [25], thereby affecting transcription of innate-immune genes and inflammatory responses. MTHFD1, a key enzyme in one-carbon metabolism, contributes to redox balance; disruption of this pathway increases ROS levels in macrophages and promotes a shift toward the pro-inflammatory M1 phenotype [26]. CDK5, although originally studied in the nervous system, has recently been shown to promote NF- κ B-related cytokine expression in macrophages and regulate M1/M2 polarization [27]. NDUFB11, an essential subunit of mitochondrial complex I, is required for proper electron transport; its dysfunction leads to impaired

electron-transfer efficiency and excessive ROS generation [28]. Enhanced mitochondrial redox stress can drive macrophages toward an M1 phenotype and promote sustained inflammation [29]. Thus, NDUFB11-related mitochondrial stress may represent an important intersection between inflammation and metabolic dysfunction in PCOS ovaries.

It should be noted that the molecular docking and MD simulations performed in this study are hypothesis-generating and are not intended to demonstrate direct physical binding between EA and POLR1C. Instead, these analyses were used to prioritize candidate molecular nodes potentially responsive to EA exposure. Future studies employing biochemical binding assays or genetic perturbation approaches will be required to determine whether EA directly interacts with or functionally regulates POLR1C.

In conclusion, this study reveals the potential toxicological mechanisms of EA in PCOS, suggesting that it may exacerbate ovarian dysfunction and metabolic imbalance by disrupting purine metabolism, impairing mitochondrial function, activating inflammatory pathways, and remodeling the immune microenvironment. These findings provide new evidence for understanding the reproductive toxicity of trans fatty acids under high-fat dietary conditions and offer potential insights for dietary management and targeted intervention.

5. Conclusion

This study systematically reveals the marked accumulation of elaidic acid in polycystic ovary syndrome and its potential metabolic toxicological mechanisms. EA may disrupt purine and energy metabolism, impair mitochondrial function, induce endoplasmic reticulum stress, and activate inflammatory signaling pathways, ultimately leading to ovarian energy imbalance and cellular injury. As a harmful dietary metabolite associated with high-fat intake, EA may serve as a key mediator linking ovarian metabolic dysfunction with immune activation. These findings provide new theoretical evidence for understanding the reproductive toxicity of trans fatty acids and for developing dietary intervention strategies in PCOS.

CRedit authorship contribution statement

Hua Ma: Conceptualization. **Li Ning:** Data curation. **Ruyue Wang:** Formal analysis. **Yating Xu:** Writing – original draft, Conceptualization. **Li Xiu:** Investigation. **Qingling Ren:** Writing – review & editing. **Hongting Zhao:** Formal analysis. **Yu Si:** Investigation.

Ethics approval and consent to participate

Not applicable.

Author contributions

Yating Xu and Li Ning designed the study, performed the experiments, analyzed the data, and drafted the manuscript. Ruyue Wang contributed to data acquisition and animal experiments. Hua Ma and Hongting Zhao assisted with methodology and data interpretation. Yu Si and Qingling Ren supervised the project, provided critical revisions, and approved the final manuscript. Yating Xu and Li Ning contributed equally to this work.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.reprotox.2026.109200.

Data availability

Data will be made available on request.

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